

Counterfactual Operator Atlas (COA)

This document formalizes **counterfactual prime theory** as a constrained dynamical system on *(representations, operators)*, with an explicit **bounded-schema rule** for the latent/adjacent-possible operators $\tilde{O}_t$. The bounded-schema rule is the central guardrail against hindsight: it makes “adjacent possible” **auditable**.

0. Aim

At each historical time-slice t , mathematicians operate within a constrained methodological world:

- **Representations**: what kinds of objects are allowed to exist and be manipulated.
- **Operators**: what kinds of inferential moves count as legitimate.
- **Selection pressures**: which moves are rewarded, legible, checkable, and prestigious.

Counterfactual history asks: *if a different but historically plausible operator/representation basis had been selected, what alternate prime-theoretic pipeline would have emerged?*

1. State variables

Fix universes:

- $\mathcal{R}$: a universe of representational primitives (e.g., residues, roots of unity, periodic functions, ideals, manifolds-as-metaphor).
- $\mathcal{O}$: a universe of operators (e.g., Euler products, orthogonality filters, generating functions, contour shifts, explicit formula heuristics, local-to-global gluing).

A **historical state** at time t is:

$$X_t := (R_t, O_t), \text{ where } R_t \subseteq \mathcal{R}, O_t \subseteq \mathcal{O}.$$

Objects admitted as “real” and manipulable at time t .

Examples (schematic):

- 18th c.: formal power series, roots of unity, congruences, finite sums, products, infinite series.
- 19th c.: analytic continuation, Riemann surfaces, complex integration, functional equations.

Inferential moves admissible at time t .

Examples (schematic):

- Expand $\prod_p (1 - p^{-s})^{-1}$ into $\sum n^{-s}$ (Euler product).
- Use roots-of-unity filters to isolate residue classes.
- Use analytic continuation/contour deformation to derive explicit formulas.

We define $\widetilde{O}_t$ as the set of **auditable** operators that are *one or a few concept-steps away* from (R_t, O_t) , using only syntactic/definitional resources available at time t .

2.1 The bounded-schema guardrail (anti-hindsight rule)

To prevent anachronism, we restrict latent moves to those expressible by **bounded-length schemas** over the current state.

(A) Schemas

Fix a formal meta-language L_t whose symbols are drawn only from:

- Object-formers permitted by R_t (e.g., sums, products, finite group operations, basic analytic primitives if present).
- Operators already in O_t (used as admissible inference rules).
- A small fixed library of **meta-constructions** allowed across all eras (e.g., substitution, composition, instantiation), treated as cognitive-neutral.

A **schema** is a finite template σ in L_t which, when instantiated with objects from R_t and operators from O_t , produces:

- either a candidate new representation $r \in \mathcal{R}$, or
- a candidate new operator $o \in \mathcal{O}$.

(B) Bounded length and bounded depth

Choose parameters (to be calibrated):

- ℓ : maximum description length (e.g., number of symbols/tokens in σ).
- d : maximum composition depth (e.g., nesting level of constructions).

A candidate o is **latent-admissible** at time t if there exists a schema $\sigma \in L_t$ such that:

1. $\text{len}(\sigma) \leq \ell$
2. $\text{depth}(\sigma) \leq d$
3. σ instantiates using only (R_t, O_t) to yield o

We then set:

$$\widetilde{O}_t := \{o, o \in \mathcal{O} : \exists \sigma \in L_t \text{ s.t. (1-3) hold}\}.$$

Interpretation: ℓ and d are a proxy for “one or two conceptual steps away,” making adjacent-possible claims checkable.

(C) Auditing rule

For each $o \in \widetilde{O}_t$, an audit must include:

- the explicit schema σ ,
- the instantiation map (which objects/operators in (R_t, O_t) are used),
- and a minimal derivation showing the operator is well-defined under R_t .

This creates a paper trail that prevents “smuggling” future abstractions.

3. Transition dynamics

Counterfactual evolution is a constrained search or stochastic process through states X_t .

3.1 Candidate generation

Define an **adjacent generator**:

$$A(X_t) := \{(R_t \cup \{r\}, O_t) : r \in \widetilde{R}_t\} \cup \{(R_t, O_t \cup \{o\}) : o \in \widetilde{O}_t\},$$

where $\widetilde{R}_t$ is defined analogously to $\widetilde{O}_t$ via bounded schemas that create new objects.

Optionally allow a bounded closure operator cl_k permitting up to k sequential adjunctions in one macro-step.

Let S_t be a **selection/incentive functional**:

$$S_t : \{\text{candidate states}\} \rightarrow \mathbb{R}.$$

It may combine:

- **Legibility** (is it expressible in L_t and explainable?),
- **Checkability** (can it be verified by computations/examples available?),
- **Problem alignment** (does it advance prestige problems of the era?),
- **Institutional fit** (publication norms, pedagogy, genre),
- **Cognitive friction** (penalize long derivations, nonstandard objects).

3.3 Dynamics (one canonical choice)

A softmax transition (conservative when β is large):

$$\mathbb{P}(X_{t+1}=Y \mid X_t) \propto \exp(\beta_t S_t(Y)) \cdot \mathbf{1}_{Y \in \mathcal{A}_t(X_t)}$$

Deterministic alternative: pick $X_{t+1} \in \arg \max S_t$ within the admissible neighborhood.

4. Counterfactual artifacts (non-vibes requirement)

Every proposed transition must emit a **mathematical artifact** legible in the era's language:

- a definition,
- a lemma/proof sketch,
- a computable heuristic,
- a conjecture,
- an explicit formula or identity.

Formally, attach an artifact map:

$$\mathcal{A}_t : (X_t \rightarrow X_{t+1}) \mapsto \{\text{artifacts}\}.$$

A counterfactual history is valid only if $\mathcal{A}_t$ is nonempty at every step.

5. Two canonical forks (templates)

5.1 Euler fork (GL(1)-style symmetry sectors)

Minimal representation adjunction: **multiplicative periodic functions** χ valued in roots of unity.

Latent operator produced by bounded schema:

$$\text{Twist}(\zeta) : \text{quad } L(s, \chi) = \prod_p (1 - \chi(p) p^{-s})^{-1}.$$

Artifacts: twisted Euler product identity, orthogonality-based residue filters, nonvanishing heuristic at $s = 1$.

5.2 Riemann fork (trace/spectrum worldview)

Minimal operator adjunction: **trace principle template**

$$\text{Prime-sum} \approx \text{Tr}(\text{operator}).$$

Artifacts: proto-explicit-formula-as-trace, spectral-bound restatement of RH ("eigenvalues lie on a critical line/band").

6. Fast validation protocol (implementation-minded)

1. **Vocabulary build:** define a small controlled list of $\mathcal{R}, \mathcal{O}$ for a narrow window (e.g., 1730–1840).
 2. ****Choose ** ℓ, d, k and document them.**
 3. **Retrodiction test:** fit S_t on early slices; test if historically adopted moves are ranked highly.
 4. **Ablation/counterfactual test:** add one adjunction (e.g., “characters as objects”) and measure the shortest-path reduction to key artifacts.
 5. **Audit trail:** every latent operator must ship with its schema σ + instantiation.
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7. What counts as success

- **Auditability:** every “adjacent possible” claim is backed by a bounded schema witness.
 - **Compression:** minimal changes in (R_t, O_t, S_t) yield large differences in reachable prime-theoretic artifacts.
 - **Predictive leverage:** the model retrodicts known near-misses and predicts plausible alternate sequences that domain experts accept as legible for the era.
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8. References

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